中药毒性预测由
异构网络
补充文件

1. 数据集

1.1. 数据集构建

为了构建中药毒性数据集，我们从权威来源汇编了中药的综合列表，如 Chp (2020) [1], HERB [2], SymMap v2 [3], TCM-Bank [4]，和 HIT 2.0 [5]。对于每种中药，我们手动从主要的科学数据库和文献库中检索并整理了与毒性相关的信息，包括 Google Scholar (https://scholar.google.com)、中国知网 (CNKI, https://www.cnki.net/)、维普 (VIP, https://www.cqvip.com/)、万方数据 (WanFang, https://www.wanfangdata.com.cn/index.html)、PubMed (https://pubmed.ncbi.nlm.nih.gov/) 和 Web of Science (WOS, https://www.webofscience.com/wos/)。所有收集的数据均经过人工筛选、注释和标准化，确保中药名称和毒性信息在各来源之间的一致性和可靠性。

我们定义了五种毒性标签：心脏毒性、肾毒性、肝毒性、系统毒性和其他毒性，这些标签是基于临床相关性和文献普遍性确定的。对于每种草药，其毒性类型的存在与否是通过手动审查收集的文献和数据库条目来确定的。我们的最终数据集包含252种草药，每种草药可能与多个毒性标签相关。总体而言，有74种草药被标记为心脏毒性，101种为肾毒性，114种为肝毒性，167种为系统毒性，150种具有其他毒性。

1.2. 数据集分析

图 S1 展示了所构建的多标签毒性数据集的统计概览。如图 S1(a) 所示，在收集的 252 种草药中，毒性标签的分布明显不平衡。观察到系统毒性和其他毒性。

Traditional Chinese Medicine Toxicity Prediction by Heterogeneous Network
Supplementary File

1. Dataset

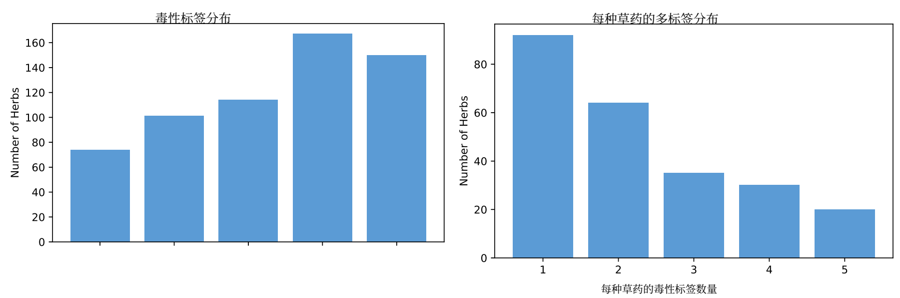
1.1. Dataset construct

To construct the herb toxicity dataset, we compiled a comprehensive list of herbs from authoritative sources such as Chp (2020) [1], HERB [2], SymMap v2 [3], TCM-Bank [4], and HIT 2.0 [5]. For each herb, toxicity-related information was manually retrieved and curated from major scientific databases and literature repositories, including Google Scholar (`https://scholar.google.com`), CNKI (China National Knowledge Infrastructure, `https://www.cnki.net/`), VIP (China Science and Technology Journal Database, `https://www.cqvip.com/`), WanFang (Wanfang Data, `https://www.wanfangdata.com.cn/index.html`), PubMed (`https://pubmed.ncbi.nlm.nih.gov/`), and WOS (Web of Science, `https://www.webofscience.com/wos/`). All collected data were manually screened, annotated, and standardized, ensuring the consistency and reliability of herb names and toxicity information across sources.

We defined five toxicity labels: cardiotoxicity, nephrotoxicity, hepatotoxicity, systemic toxicity, and other toxicity, based on clinical relevance and literature prevalence. For each herb, the presence or absence of a toxicity type was determined by manually reviewing the collected literature and database entries. Our final dataset consists of 252 herbs, each potentially associated with more than one toxicity labels. In total, 74 herbs are labeled as cardiotoxic, 101 as nephrotoxic, 114 as hepatotoxic, 167 as systemically toxic, and 150 as having other toxicities.

1.2. Dataset analysis

Figure S1 presents a statistical overview of the constructed multi-label toxicity dataset. As depicted in Figure S1(a), the distribution of toxic labels among 252 collected herbs is notably imbalanced. Systemic toxicity and other toxicity are observed



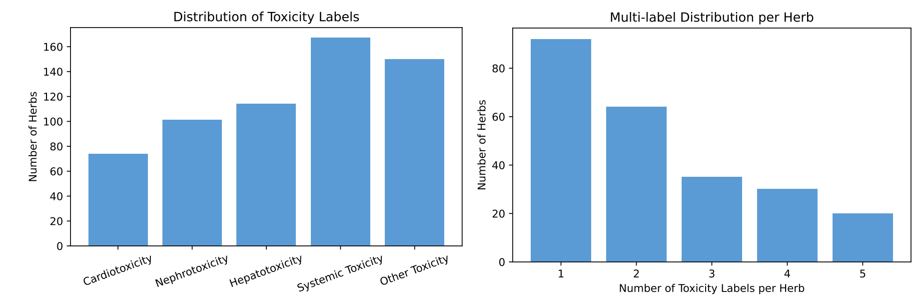
(a) 药草中毒性标签的分布 (b) 每种药草分配毒性标签数量的分布

图 S1：多标签毒性数据集的统计概览。(a) 收集的 252 种草药中每种毒性标签的频率，突出显示不同毒性类型之间的不平衡分布。(b) 每种草药的毒性标签分布，显示数据集中多标签毒性的普遍存在。

最常见的标签分别出现在167种和150种草药中，而心脏毒性相对罕见，仅在74种草药中出现。肾毒性和肝毒性分别出现在101种和114种草药中。这种分布不均表明某些毒性类型更常被报告或研究，这可能给模型训练和评估带来挑战。

图 S1(b) 进一步凸显了数据集的显著多标签特性。虽然部分草药仅标注了单一毒性标签，但大多数草药与两种或以上的毒性类型相关，有些草药甚至被分配到多达五个不同的标签。这种多标签标注的普遍存在反映了草药毒性的固有复杂性，并强调了预测模型有效捕捉多种毒性结果之间相互作用和依赖关系的必要性。多标签实例的占主导地位进一步增加了预测任务的挑战，需要能够应对这种复杂性的模型方法。

图S2显示了我们数据集中有毒标签的共现热图。结果表明，某些毒性类型往往以较高的频率共现。值得注意的是，肝毒性和肾毒性在76种草药中被共同标注，表明这两种毒性类型之间存在较强的关联。此外，



(a) Distribution of toxicity labels among the herb dataset. (b) Distribution of the number of toxicity labels assigned to each herb.

Figure S1: Statistical overview of the multi-label toxicity dataset. (a) The frequency of each toxicity label among the collected 252 herbs, highlighting the imbalanced distribution across different toxicity types. (b) The distribution of toxic labels per herb, demonstrating the prevalence of multi-label toxicity within the dataset.

to be the most frequently assigned labels, occurring in 167 and 150 herbs, respectively, whereas cardiotoxicity is comparatively rare, with only 74 herbs. Nephrotoxicity and hepatotoxicity are present in 101 and 114 herbs, respectively. This uneven distribution suggests that certain toxicity types are more commonly reported or studied, which may introduce challenges for model training and evaluation.

Figure S1(b) further highlights the pronounced multi-label characteristics of the dataset. While a portion of herbs are annotated with a single toxic label, the majority are associated with two or more toxic types, with some herbs assigned up to five distinct labels. This prevalence of multi-label annotations reflects the inherent complexity of herb toxicity and underscores the necessity for predictive models to effectively capture the interactions and dependencies among multiple toxicity outcomes. The dominance of multi-label instances further increases the challenge of the prediction task, demanding approaches that are capable of addressing such complexity.

Figure S2 displays the co-occurrence heatmap of toxic labels within our dataset. The results demonstrate that certain toxicity types tend to co-occur at relatively high frequencies. Notably, hepatotoxicity and nephrotoxicity are jointly annotated in 76 herbs, indicating a strong association between these two toxicity types. Additionally,

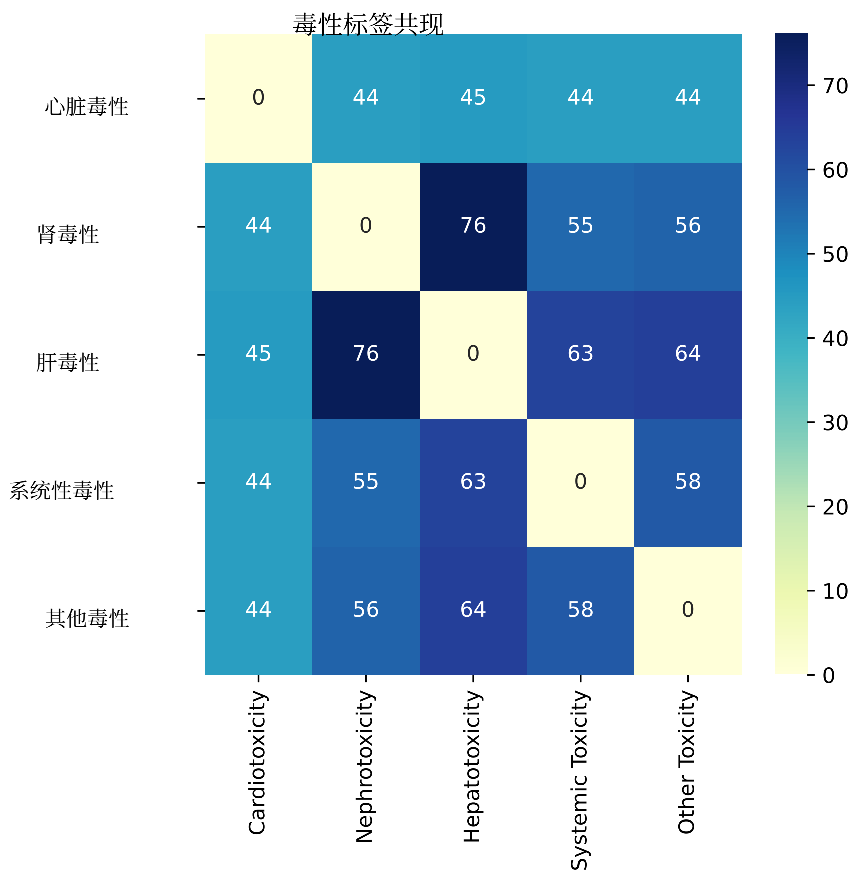


图 S2：草药毒性标签的共现热图。

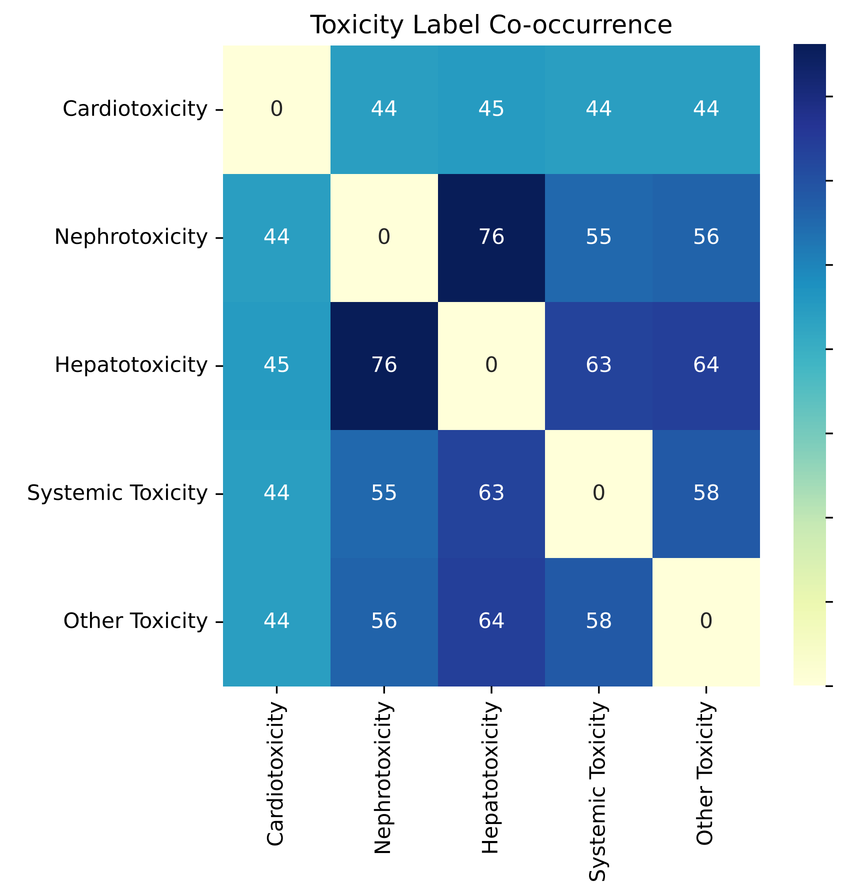


Figure S2: Co-occurrence heatmap of toxic labels of herbs.

系统性毒性常与肾毒性（55种草药）及其他毒性（58种草药）同时存在，这反映了这些类别之间潜在的相互依赖或共同的风险因素。这种共现模式可能反映了潜在的生物学相关性或多重毒性反应背后的共同机制。频繁的共现现象，加上标签不平衡和普遍的多标签关联，突显了数据集的固有复杂性，并可能增加开发准确且具有可推广性的草药毒性预测模型的难度。

2. 消融研究

为了进一步研究 HerbToxNet 的贡献因素，我们引入了五种变体：(i) w/o HAN 直接将药材的属性和功效数据作为特征使用，而不构建异构图和元路径，但结合对比学习和加权标签融合策略进行预测。(ii) w/o PE 在异构图中随机初始化药材节点特征，而不是使用药理属性和功效数据作为初始特征。(iii) w/o CL 在训练阶段移除带有动态系数的对比学习模块。(iv) w/o $\boldsymbol{\delta}$ 用毒性标签相似度 β_{ij} 替代动态系数 δ_{ij} 。(v) w/o WLF 仅使用 MLP 进行预测，不使用加权标签融合策略。

表S1显示了HerbToxNet及其退化变体的结果。总体而言，HerbToxNet在所有评估指标上显著优于其变体，凸显了各设计模块在完整模型中的重要性。具体来说：(i) HerbToxNet明显优于去掉HAN的模型，这表明HAN能够有效建模草药、成分和靶点之间的复杂关系，通过元路径学习更具判别性的草药表征，从而为毒性预测提供更可靠的基础。(ii) HerbToxNet比去掉PE的模型表现更好，这表明药理属性和功效数据尽管存在固有的不完整性和不足性，但仍为草药毒性预测提供了有价值的信息。这些数据为模型补充了领域知识，帮助模型更好地理解草药的毒性特征。(iii) HerbToxNet优于去掉CL的模型，这是因为⋯

systemic toxicity frequently co-exists with nephrotoxicity (55 herbs) and other toxicity (58 herbs), reflecting potential interdependencies or shared risk factors among these categories. Such co-occurrence patterns may reflect potential biological correlations or shared mechanisms underlying multiple toxic responses. The presence of frequent co-occurrences, together with label imbalance and prevalent multi-label associations, highlights the inherent complexity of the dataset and may increase the challenges associated with developing accurate and generalizable toxicity prediction models of herbs.

2. Ablation study

To further study the contribution factors of `HerbToxNet`, we introduce five variants: (i) **w/o HAN** directly uses the properties and efficacy data of herb as features without constructing heterogeneous graph and the meta-path, but combined with contrastive learning and weighted label fusion strategies for prediction. (ii) **w/o PE** randomly initializes the herb node features in the heterogeneous graph instead of using pharmacological property and efficacy data as initial features. (iii) **w/o CL** removes the contrastive learning module with dynamic coefficients in the training phase. (iv) **w/o $\boldsymbol{\delta}$** replaces the dynamic coefficient δ_{ij} with the toxicity label similarity β_{ij}. (v) **w/o WLF** only uses the MLP for prediction, without incorporating the weighted label fusion strategy.

Table S1 shows the results of `HerbToxNet` and its degenerated variants. Overall, `HerbToxNet` significantly outperforms its variants across all evaluation metrics, highlighting the importance of each designed module in the full model. Specifically: (i) `HerbToxNet` clearly outperforms w/o HAN, this indicates HAN effectively models the complex relationships between herbs, ingredients, and targets, learns more discriminative herb representations by meta-path, thus providing a more reliable foundation for toxicity prediction. (ii) `HerbToxNet` gains better results than w/o PE, suggesting that pharmacological property and efficacy data provide valuable information for herb toxicity prediction, despite their inherent incomplete and insufficiency. These data supplement the model with domain knowledge, helping the model to better understand the toxic characteristics of herbs. (iii) `HerbToxNet` is superior to w/o CL, owing to the con-

对比学习将具有相似毒性标签的药材拉近，将不同的药材拉开，这显著提高了模型的鲁棒性和准确性。(iv) HerbToxNet 的表现优于没有 δ 的模型，这验证了超参数 t 可以调整 β，并允许对药材之间的语义相似性进行更灵活的量化，从而有利于对比学习。(v) HerbToxNet 优于没有 WLF 的模型，这表明加权标签融合策略能够有效利用来自其他相似药材的知识，通过融合相似药材的毒性标签并缓解数据稀缺问题，进一步提高预测准确性。

我们观察到，在所有指标中，去掉对比学习模块（w/o CL）的性能最差，这凸显了对比学习模块在增强草药判别表示中的关键作用。去掉HAN的变体（w/o HAN）的表现仅略好一些，强调了元路径机制在捕捉草药、成分和靶点之间复杂语义关联中的重要性。相比之下，尽管去掉PE和去掉WLF（w/o PE 和 w/o WLF）获得了相对较高的分数，其性能仍显著低于HerbToxNet，这表明药理属性和功效信息的整合，以及加权标签融合策略，对模型预测具有重要贡献。去掉 δ 的性能介于去掉HAN和去掉PE之间，这表明动态系数调整对调节对比学习过程有不可忽视的影响，尽管程度较小。总体而言，这些结果表明HerbTo中每个组件的重要性。

表 S1： `HerbToxNet`及其变体的预测结果。

	宏观 F1	微F1	平均AUC	1–一个错误
无HAN	0.4934±0.0357	0.5352±0.0218	0.5003±0.0245	0.5335±0.0378
无体育课	0.4766±0.0312	0.5435±0.0304	0.4854±0.0153	0.5475±0.0370
无 CL	0.4366±0.0292	0.5305±0.0262	0.4851±0.0168	0.5500±0.0177
无 δ	0.5087±0.0342	0.5411±0.0216	0.5252±0.0206	0.5486±0.0152
无 WLF	0.5269±0.0317	0.6131±0.0233	0.5363±0.0313	0.5536±0.0511
`HerbToxNet`	**0.5652± 0.0161**	**0.6247±0.0172**	**0.6003± 0.0141**	**0.5587±0.0485**

注意：最佳结果以粗体显示，统计显著性通过 95% 水平的学生 t 检验进行检验。

trastive learning which pulls together herbs with similar toxic labels and pushes apart dissimilar ones, this significantly enhances the robustness and accuracy of the model. (iv) `HerbToxNet` outperforms w/o δ, which validates that the hyperparameter t can adjust β and allow for more flexible quantification of semantic similarity between herbs, thereby rewarding contrastive learning. (v) `HerbToxNet` is better than w/o WLF, suggesting the weighted label fusion strategy effectively leverages knowledge from other similar herbs, further improving prediction accuracy by fusing the toxic labels from similar herbs and by alleviating the scarce data issue.

We observe that the performance of w/o CL is the poorest across all metrics, underscoring the critical role of the contrastive learning module in enhancing discriminative representation of herbs. The variant w/o HAN performs only slightly better, highlighting the importance of the meta-path mechanism in capturing the complex semantic associations among herbs, ingredients, and targets. In contrast, although w/o PE and w/o WLF achieve relatively higher scores, their performance remains significantly inferior to `HerbToxNet`, suggesting that the integration of pharmacological property and efficacy information, along with the weighted label fusion strategy, substantially contributes to model prediction. The performance of w/o δ falls between that of w/o HAN and w/o PE, indicating that dynamic coefficient adjustment has a non-negligible impact on modulating the contrastive learning process, albeit to a lesser extent. Overall, these results demonstrate that each component in `HerbToxNet` is important, and their synergy is crucial to achieve the state-of-the-art performance.

Table S1: Prediction results of `HerbToxNet` and its variants.

	Macro-F1	Micro-F1	AvgAUC	1−OneError
w/o HAN	0.4934±0.0357	0.5352±0.0218	0.5003±0.0245	0.5335±0.0378
w/o PE	0.4766±0.0312	0.5435±0.0304	0.4854±0.0153	0.5475±0.0370
w/o CL	0.4366±0.0292	0.5305±0.0262	0.4851±0.0168	0.5500±0.0177
w/o δ	0.5087±0.0342	0.5411±0.0216	0.5252±0.0206	0.5486±0.0152
w/o WLF	0.5269±0.0317	0.6131±0.0233	0.5363±0.0313	0.5536±0.0511
`HerbToxNet`	**0.5652± 0.0161**	**0.6247±0.0172**	**0.6003± 0.0141**	**0.5587±0.0485**

Note: The best results are highlighted in **bold font**, with statistical significance checked by student t-test at 95% level.

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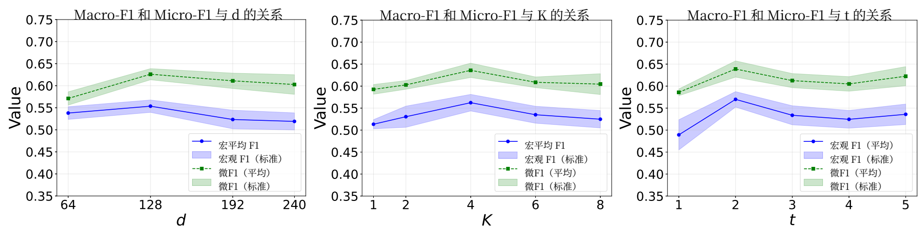
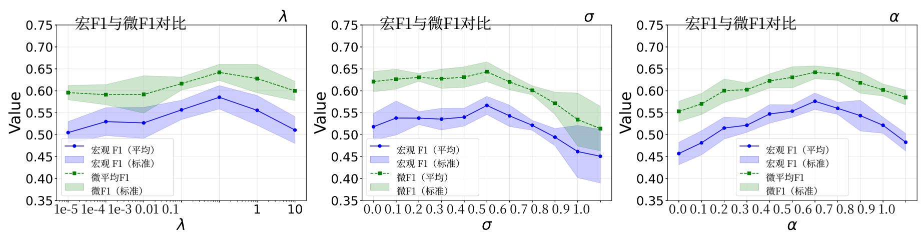


(a) 节点嵌入维度 d (b) 注意力头编号 K (c) 调整参数 t



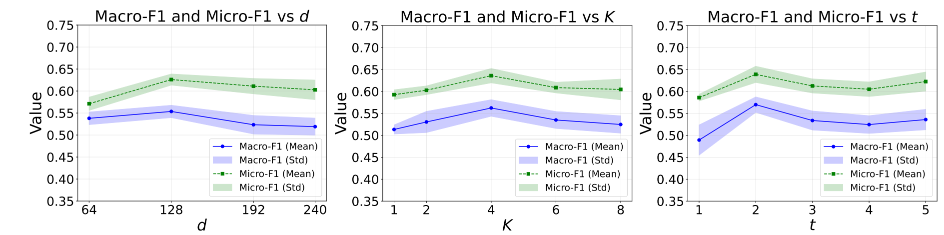
(d) 损失函数平衡参数 λ (e) 加权标签融合阈值 σ (f) 融合比例参数 α 乘

图 S3：参数敏感性分析结果。(a)、(b)、(c) 和 (d) 分别展示了 `HerbToxNet` 在不同节点嵌入维度 d、注意力头数 K、调整参数 t 以及损失函数平衡参数 λ 下的 Macro-F1 和 Micro-F1；(e) 和 (f) 分别展示了 `HerbToxNet` 在不同加权标签融合阈值 σ 和融合比例参数 α 下的 Macro-F1 和 Micro-F1。

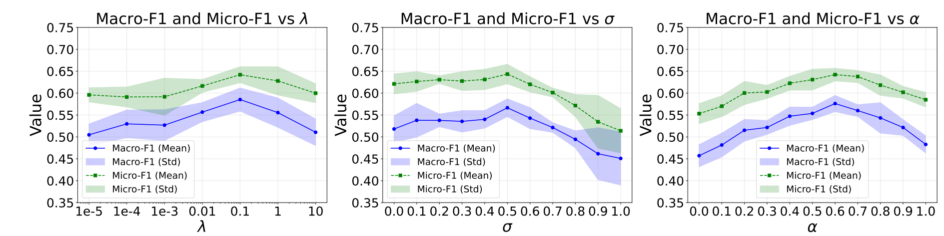
3. 参数敏感性分析

HerbToxNet 是用 Python 编程语言实现的。在模型训练中，我们使用了一块安装在 Ubuntu 22.04.1 LTS 机器上的 Nvidia GeForce RTX 3090 (24G) GPU，机器配备 Intel Xeon Gold 6248R CPU 和 512GB 内存。CUDA 版本为 11.7，PyTorch 版本为 1.13.1。

在此实验环境中，我们系统地研究了几个重要参数的影响：（i）d 决定了异质图中节点（如草药、成分和靶点）的表示维度；（ii）K 表示 HAN 中注意力头的数量，它使模型能够在不同位置联合关注来自不同表示子空间的信息；（iii）对比学习动态系数调节参数 t 用于调节毒性



(a) node embedding dimension d (b) attention head number K (c) adjustment parameter t



(d) loss function balancing parameter λ (e) weighted label fusion threshold σ (f) fusion proportion parameter α

Figure S3: Results of parameter sensitivity analysis. (a), (b), (c) and (d) show the Macro-F1 and Micro-F1 of `HerbToxNet` under different input values of node embedding dimension d, head number K, adjustment parameter t, and loss function balancing parameter λ, respectively; (e) and (f) show the Macro-F1 and Micro-F1 of `HerbToxNet` under different input values of the weighted label fusion threshold σ and fusion proportion parameter α, respectively.

3. Parameter sensitivity analysis

`HerbToxNet` is implemented by Python program language. For the model training, We used an Nvidia GeForce RTX 3090 (24G) GPU that hosted on a Ubuntu 22.04.1 LTS machine with Intel Xeon Gold 6248R CPUs and 512GB memory. The version of CUDA is 11.7 and that of PyTorch is 1.13.1.

In this experimental environment, we systematically investigated the impact of several important parameters: (i) d determines the representation dimensions of nodes such as herbs, ingredients, and targets in the heterogeneous graph; (ii) K denotes the number of attention heads in the HAN, which allows the model to jointly attend to information from different representation subspaces at different positions. (iii) Contrastive learning dynamic coefficient adjustment parameter t is used to adjust the toxic-

标签相似度 β_{ij}，它会影响动态系数 δ_ij 的计算。(iv) 损失函数平衡参数 λ 平衡对比学习损失 $\mathcal{L}^{con}$ 与非对称分类损失 $\mathcal{L}^{asl}$。(v) 加权标签融合阈值 σ 选择与其相似度较高的相似药材以补充标签。(vi) 融合比例参数 α 控制来自 MLP 和加权标签融合的预测权重。我们测试 d 在 $\{64, 128, 192, 240\}$ 范围内，K 在 $\{1, 2, 4, 6, 8\}$ 范围内，t 在 $\{1, 2, 3, 4, 5\}$ 范围内，λ 在 $\{10^{-5}, 10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 10^{0}, 10\}$ 范围内，σ 和 α 都在 {0, 0 范围内。1、0.2、0.3、0.4、0.5、0.6、0.7、0.8、0.9、1。0} 在我们的数据集上。当测试每个参数时，其他参数固定在其最优值，结果如图 S3 所示。

节点嵌入维度 d 起着关键作用，HerbToxNet 在 d = 128 时达到最佳性能，这表明该嵌入维度能够有效捕捉异构网络中草药、成分和靶点之间的复杂关联。相比之下，当 d = 64 时，模型的 Macro-F1 分数显著下降，说明较低的嵌入维度限制了模型捕捉节点间复杂关系的能力。然而，当 d ≥ 128 时，其性能甚至低于 d = 64 时的表现。这表明过大的嵌入维度可能会引入噪声，同时造成显著更高的计算成本。因此，我们选择 d = 128 作为最佳嵌入维度，以确保节点的有效表示。

注意力头的数量 K 对 HerbToxNet 的性能有显著影响。实验结果表明，当 K = 4 时，模型达到最佳性能，这强调了使用多头注意力从异质节点交互中捕获多样语义模式的有效性。当 K = 1 时，单头注意力机制缺乏足够的能力来区分多样的语义模式，导致预测性能明显下降。当 K > 4 时，性能开始下降，这可能是由于过多的注意力头分散了信息信号，从而削弱了整体特征表示。因此，我们选择 K = 4 作为最佳配置，以确保足够的表现力而不引入噪声或不稳定性。

动态系数调整参数 t 在优化中起着关键作用

ity label similarity β_{ij}, which affects the calculation of the dynamic coefficient δ_{ij}. (iv) Loss function balancing parameter λ balances the contrastive learning loss $\mathcal{L}^{con}$ and the asymmetric classification loss $\mathcal{L}^{asl}$. (v) Weighted label fusion threshold σ selects similar herbs with higher similarity to supplement the labels. (vi) Fusion proportion parameter α controls the weights of predictions from MLP and weighted label fusion. We test d within $\{64, 128, 192, 240\}$, K within $\{1, 2, 4, 6, 8\}$, t within $\{1, 2, 3, 4, 5\}$, λ within $\{10^{-5}, 10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 10^{0},$
$10\}$, both σ and α within $\{0, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1.0\}$ on our dataset. When testing each parameter, the other parameters are fixed at their optimal values and the results are shown in Figure S3.

The node embedding dimension d plays a pivotal role, `HerbToxNet` achieves the optimal performance when $d = 128$, suggesting that this embedding dimension effectively captures the intricate associations among herbs, ingredients, and targets within the heterogeneous network. In contrast, when $d = 64$, the model exhibits a significantly lower Macro-F1 score, indicating that a lower embedding dimension limits the model's ability to capture the complex relationships among nodes. However, when $d \geq 128$, its performance even falls below that observed at $d = 64$. This suggests that a excessively large embedding dimension may introduce noise, while also incurring substantially higher computational costs. Consequently, we select $d = 128$ as the optimal embedding dimension to ensure effective representation of nodes.

The number of attention heads K has a significant impact on the performance of `HerbToxNet`. Experimental results demonstrate that the model achieves its best performance when $K = 4$, emphasizing the effectiveness of employing multi-head attention to capture diverse semantic patterns from heterogeneous node interactions. When $K = 1$, the single-head attention mechanism lacks sufficient capacity to distinguish diverse semantic patterns, resulting in a noticeable drop in prediction performance. As $K > 4$, the performance starts to decline, likely due to excessive attention heads dispersing informative signals, thereby weakening the overall feature representation. Therefore, we choose $K = 4$ as the optimal configuration to ensure sufficient expressiveness without introducing noise or instability.

The dynamic coefficient adjustment parameter t plays a critical role in the opti-

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对比学习的优化过程。实验结果表明，当 t ＝ 2 时，模型达到最佳性能。在此值下，模型有效地平衡了对高相似样本的增强和对低相似样本的惩罚。当 t ＝ 1 时，动态系数降低到 δ_{ij}，这不足以增强高相似样本的贡献。这一限制阻碍了模型清晰地区分不同样本的相似性。相反，t ＞ 4 对低相似样本的惩罚过度。虽然这一调整进一步强调了高相似样本，但也导致模型过度关注这些样本，可能忽略其他有价值的信息。这种过度强调可能导致整体模型性能下降。因此，我们设置 t ＝ 2 来增强高相似样本的贡献。

平衡参数 λ 在模型训练过程中起着关键作用。实验结果表明，当 $\lambda = 0.1$ 时，模型能够达到最佳性能，因为该设置在对比学习损失和多标签分类损失之间建立了合理的平衡。随着 λ 的增加，模型性能最初有所提升，但随后下降，这表明对比学习损失的过度强调会削弱多标签分类损失在最终分类任务中的优化效果。相反，当对比学习损失的贡献不足时，模型无法充分利用样本之间的相似性信息。因此，我们将 $\lambda = 0.1$ 设置为有效利用对比学习策略，同时兼顾最终分类目标。

加权标签融合阈值 σ 在检索过程中对筛选相似邻近样本起着关键作用。实验结果表明，当 $\sigma = 0.5$ 时，模型性能达到最佳。具体来说，当 σ 过小时，模型往往会包含过多的噪声样本，这导致许多与目标草药关联较弱的有害标签被纳入，从而干扰模型的学习过程。相反，当 σ 过大时，相似邻近样本的数量会显著减少，限制了模型在训练过程中有效利用这些邻近样本信息的能力。因此，我们将 $\sigma = 0.5$ 设定为选择相关相似样本并过滤噪声的相似性阈值。

融合比例参数 α 决定了目标的相对贡献

mization process of contrastive learning. Experimental results indicate that the model achieves optimal performance when $t = 2$. At this value, the model effectively balances the amplification of high-similar samples and the penalization of low-similar ones. When $t = 1$, the dynamic coefficient reduces to δ_{ij}, which insufficiently amplifies the contributions of high-similar samples. This limitation hinders the model's ability to clearly distinguish between the similarities of different samples. Conversely, $t > 4$ excessively penalizes low-similar samples. While this adjustment further emphasizes high-similar samples, it causes the model to disproportionately focus on these samples, potentially neglecting other valuable information. This overemphasis can lead to a decline in overall model performance. As such, we set $t = 2$ to enhance the contributions of high-similar samples.

The balance parameter λ plays a pivotal role in the model training process. Experimental results indicate that the model achieves optimal performance when $\lambda = 0.1$, as this setting establishes a reasonable balance between the contrastive learning loss and the multi-label classification loss. As λ increases, the model's performance initially improves but subsequently declines, suggesting that an excessive emphasis on the contrastive learning loss diminishes the optimization effect of the multi-label classification loss on the final classification task. Conversely, when the contribution of the contrastive learning loss is insufficient, the model fails to fully leverage the similarity information among samples. Such, we set $\lambda = 0.1$ to effectively utilize the contrastive learning strategy while simultaneously addressing the final classification objective.

The weighted label fusion threshold σ plays a critical role in filtering similar neighbor samples during the retrieval process. Experimental results show that the model achieves optimal performance when $\sigma = 0.5$. Specifically, when σ is too small, the model tends to include an excessive number of noisy samples. This results in the inclusion of many toxic labels with weak associations to the target herb, thereby disrupting the model's learning process. Conversely, when σ is too large, the number of similar neighbor samples is significantly reduced, limiting the model's ability to effectively leverage information from these neighbors during training. Therefore, we set $\sigma = 0.5$ as the similarity threshold to select relevant similar samples and filter out noise.

The fusion proportion parameter α determines the relative contribution of the target

在加权标签融合策略中，样本的标签及其相似样本的标签。实验结果表明，当 α =0.6 时，模型性能达到最佳。这表明适度增加邻近样本有毒标签的权重可以有效增强模型的预测能力。当 $\alpha < 0.6$ 时，邻近草药的有限影响阻碍了有效的标签融合。同时，$\alpha > 0.6$ 会导致模型过度依赖邻居，从而削弱目标样本自身的表示能力并降低性能。因此，我们将 $\alpha = 0.6$ 设定为在目标草药及其邻近草药诱导的有毒标签之间取得平衡。

基于参数敏感性分析，我们将参数值设置如下：$d = 128$，$K = 4$，$t = 2$，$\lambda = 0.1$，$\sigma = 0.5$，和 $\alpha = 0.6$。我们方法的一般训练相关参数设置，以及所有基线方法的参数设置，汇总如表 S2 所示。

表 S2： `HerbToxNet` 及所有基线方法的参数设置。

	学习率	训练轮次	输入大小	辍学率	优化器
ZF-LightGBM	0.01	100	252×364	–	–
QSAR-人工神经网络	0.001	200	252×364	–	Adam
QSAR-SVM	–	直到收敛	252×364	–	内部
EI-MS-RF	–	300	252×364	–	–
深度药物性肝损伤	0.0001	100	252×364	–	Adam
R-E-GA	–	150	252×364	–	–
TIRP网络	0.001	200	252×364	–	Adam
ML-PRDF	–	–	252×364	–	–
MulGRaL	0.001	200	252×364	0.3	Adam
MLC-CNN	0.001	100	252×364	0.5	Adam
`HerbToxNet`	0.001	200	252×364	0.2	Adam

注意：“–”表示该参数不适用于相应的方法。

4. 草药的毒性预测

在本案例研究中，我们使用 HerbToxNet 来预测和分析典型草药的毒性：柴胡、何首乌、巴豆、附子和蓖麻子在现实场景中的情况。柴胡和何首乌因其潜在的肝毒性而被广泛认可。

sample's labels and those of its similar samples in the weighted label fusion strategy. Experimental results indicate that the model achieves optimal performance when $\alpha = 0.6$. This suggests that moderately increasing the weight for toxic labels of neighbor samples effectively enhance the model's predictive capability. When $\alpha < 0.6$, the limited influence of neighbor herbs hinders effective label fusion. Meanwhile, $\alpha > 0.6$ causes the model to overly depend on neighbors, weakening the target sample's own representation and reducing the performance. Thus, we set $\alpha = 0.6$ to balance the toxic labels induced from the target herb and its neighbor herbs.

Based on the parameter sensitivity analysis, we set the parameter values as follows: $d = 128$, $K = 4$, $t = 2$, $\lambda = 0.1$, $\sigma = 0.5$, and $\alpha = 0.6$. The general training-related parameter settings of our method, as well as those of all baseline methods, are summarized in Table S2.

Table S2: Parameter settings of `HerbToxNet` and all baseline methods.

	Learning rate	Training epoch	Input size	Dropout rate	Optiminzer
ZF-LightGBM	0.01	100	252×364	–	–
QSAR-ANN	0.001	200	252×364	–	Adam
QSAR-SVM	–	Until convergence	252×364	–	Internal
EI-MS-RF	–	300	252×364	–	–
DeepDILI	0.0001	100	252×364	–	Adam
R-E-GA	–	150	252×364	–	–
TIRPnet	0.001	200	252×364	–	Adam
ML-PRDF	–	–	252×364	–	–
MulGRaL	0.001	200	252×364	0.3	Adam
MLC-CNN	0.001	100	252×364	0.5	Adam
`HerbToxNet`	0.001	200	252×364	0.2	Adam

Note: "–" indicates that the parameter is not applicable to the corresponding method.

4. Toxicity prediction of herbs

In this case study, we use `HerbToxNet` to predict and analyze the toxicity of canonical herbs: *Chai Hu*, *He Shou Wu*, *Ba Dou*, *Fu Zi*, and *Bi Ma Zi* in real-world scenarios. *Chai Hu* and *He Shou Wu* are widely recognized for their potential hepatotoxicity

并且纳入以评估模型在预测肝脏相关不良反应方面的性能。巴豆和附子分别被归类为极“寒”与“热”药材，旨在探讨它们与肝、肾和心脏系统的毒性相关性。此外，我们还加入了比麻子，该药材在传统理论中被归类为“平”（性平，一般认为无毒），以研究模型预测是否与传统观点一致，即性平药材基本安全。在实验设置中，除了这五味药材外，原始数据集中的所有其他药物作为训练集，而这五味药材作为单独的测试集用于最终预测。此设置有效评估了HerbToxNet在完全未知样本上的泛化能力。随后将预测的毒性与已知毒性标签进行比较，并从PubMed检索相关文献以验证是否

4.1. *HerbToxNet* 能够识别典型肝毒性草药的毒性

如表 S3 所示，HerbToxNet 能够准确预测柴胡的三种主要毒性：肝毒性、肾毒性和全身毒性。柴胡中活性成分的主要类别包括柴胡皂苷、挥发油、黄酮类和多糖，这些成分既负责其药理作用，也负责其毒理作用。研究显示，柴胡皂苷可引起肝毒性，包括肝酶升高和肝细胞坏死，这可能是由于抑制肝脏药物代谢酶，尤其是在肝功能受损的个体中 [6]。预测的全身毒性可归因于挥发油和皂苷引起的黏膜刺激，从而导致恶心、呕吐或腹泻等胃肠道症状 [6]。关于肾毒性，长期给予从柴胡中通过乙醇洗脱纯化的总皂苷一定剂量，已显示在大鼠中会造成显著的累积毒性，其主要靶 organ

and included to assess the model's performance in predicting liver-related adverse effects. *Ba Dou* and *Fu Zi*, classified as extremely “cold” and “hot” herbs respectively, are examined to explore their toxicity associations with liver, kidney, and cardiac systems. Additionally, we incorporated the herb *Bi Ma Zi*, which was classified as “ping” (neutral in nature and generally considered non-toxic) in traditional theories, to study whether the model predictions are consistent with the traditional view that neutral herbs are essentially safe. In the experimental setup, besides these five herbs, all other herbs in the original dataset are used as the training set, while the five selected herbs are treated as a separate test set for final prediction. This setup effectively evaluates the generalization capability of `HerbToxNet` on completely unknown samples. The predicted toxicities are then compared with known toxic labels, and relevant literature is retrieved from PubMed to verify whether the predictions align with current research findings. This experiment further assesses the scientific validity and authenticity of `HerbToxNet`.

4.1. `HerbToxNet` *can identify toxicity of canonical hepatotoxic herbs*

As shown in Table S3, `HerbToxNet` accurately predicts three major toxicities of *Chai Hu*: hepatotoxicity, nephrotoxicity, and systemic toxicity. The primary categories of active ingredients in *Chai Hu* include saikosaponins, volatile oils, flavonoids, and polysaccharides, which are responsible for both its pharmacological and toxicological properties. Saikosaponins have been shown to induce hepatotoxicity, including elevated liver enzymes and hepatocyte necrosis, likely due to inhibition of hepatic drug-metabolizing enzymes, especially in individuals with impaired liver function [6]. The predicted systemic toxicity can be attributed to mucosal irritation caused by volatile oils and saponins, leading to gastrointestinal symptoms such as nausea, vomiting, or diarrhea [6]. For nephrotoxicity, long-term administration of a certain dose of total saikosaponins purified by ethanol elution from *Chai Hu* has been shown to cause significant cumulative toxicity in rats, with the primary target organs of toxic damage being the liver and kidneys [7]. For *He Shou Wu*, `HerbToxNet` also produces accurate toxicity predictions, including hepatotoxicity and nephrotoxicity, along with a possible systemic toxicity signal. *He Shou Wu* contains ingredient categories such as

蒽醌类、白藜芦醇、 多糖和磷脂。尽管它因其抗氧化和护肝作用而被广泛使用，但过量或长期使用与严重的肝损伤有关，包括肝炎、肝功能衰竭，甚至需要肝脏移植 [8]。预测的肾毒性与已有证据一致，即蒽醌类可在动物模型中引起肾小管损伤和间质纤维化 [9]。虽然直接的系统毒性证据有限，但过敏反应的报告，包括皮疹和瘙痒等免疫介导的症状，表明该模型能够捕捉低频或情境依赖的系统毒性效应 [10]。

4.2. *HerbToxNet* 可以揭示极寒或极热草药的毒性风险

为了进一步研究极端药性与毒性之间的关系，我们分别分析了代表性极“寒”药和极“热”药巴豆和附子。根据传统理论，这类药物具有强烈的生物活性，可能导致器官功能过度刺激或紊乱，从而增加其毒性潜力。根据表S3所示的结果，HerbToxNet 成功识别出巴豆的多种毒性，包括肾毒性、肝毒性、系统毒性（消化系统毒性）及其他毒性（遗传毒性）。巴豆的活性成分苦楝油和珀尔酯具有显著的毒性作用。巴豆的肝毒性主要归因于其成分对肝细胞膜和线粒体的直接损伤，可能导致细胞坏死和肝酶升高。消化系统毒性则源于苦楝油的强烈刺激作用，可能会损害消化系统功能……

同样地，HerbToxNet 能准确预测附子的心脏毒性。附子的主要毒性成分是乌头碱类生物碱，过量或加工不当时能够迅速作用于心脏钠通道，导致心律失常、心动过缓，甚至猝死 [14]。总体而言，这些结果表明 HerbToxNet 可以

anthraquinones, stilbenes, polysaccharides, and phospholipids. Although it is widely used for its antioxidant and hepatoprotective effects, excessive or prolonged use are associated with severe liver injury, including hepatitis, liver failure, and even liver transplantation [8]. The predicted nephrotoxicity aligns with evidence that anthraquinones can cause tubular damage and interstitial fibrosis in animal models [9]. Although direct evidence for systemic toxicity is limited, reports of allergic reactions, including immune-mediated symptoms such as rash and pruritus, suggest the model can capture low-frequency or context-dependent systemic toxic effects [10].

4.2. `HerbToxNet` *can reveal toxicity risks in extremely cold or hot herbs*

To further investigate the relationship between extreme herb properties and toxicity, we analyzed representative extremely “cold” and “hot” herbs *Ba Dou* and *Fu Zi*, respectively. According to traditional theory, such herbs exhibit potent bioactivity and may cause overstimulation or disruption of organ function, contributing to their toxic potential. According to the results shown in Table S3, `HerbToxNet` successfully identified multiple toxicities of *Ba Dou*, including nephrotoxicity, hepatotoxicity, systemic toxicity (digestive system toxicity) and other toxicity (genetic toxicity). The active ingredients *croton oil* and *phorbol esters* of *Ba Dou* possess significant toxic effects. The hepatotoxicity of *Ba Dou* is primarily attributed to the direct damage of its ingredients to hepatocyte membranes and mitochondria, potentially resulting in cell necrosis and elevated liver enzymes [11]. The digestive system toxicity stems from the strong irritant effects of *croton oil*, which can compromise intestinal barrier integrity and trigger mucosal inflammation [12]. Its genetic toxicity is attributed to *phorbol esters*, which promote DNA damage and chromosomal aberrations via protein kinase C activation, contributing to carcinogenic processes [11]. Although literature on its nephrotoxicity is relatively limited, animal studies have shown that high doses of *Ba Dou* cause renal congestion and tubular degeneration, indicating kidney damage [13].

Similarly, `HerbToxNet` accurately predicts the cardiotoxicity of *Fu Zi*. The primary toxic ingredient of *Fu Zi* is *aconitine alkaloids*, which can rapidly act on cardiac sodium channels when overdosed or improperly processed, leading to arrhythmia, bradycardia, or even sudden cardiac death [14]. Overall, these results indicate that `HerbToxNet` can

对极“寒”或极“热”的草药进行准确可靠的毒性预测。

表 S3：柴胡、何首乌、巴豆、附子、毕麻子的毒性预测结果

草药名称	Rank	毒性	预测	已知	PubMed 编号
柴胡	1	肝毒性	√	√	21749378
	2	肾毒性	√	√	34309413
	3	系统性毒性	√	√	21749378
	4	其他毒性	×	×	-
	5	心脏毒性	×	×	-
何首乌	1	肝毒性	√	√	25648693
	2	**系统性毒性**	√	×	**19873791**
	3	肾毒性	√	√	32426606
	4	其他毒性	×	×	-
	5	心脏毒性	×	×	-
巴豆	1	其他毒性	√	√	35619875
	2	肾毒性	√	√	32943947
	3	肝毒性	√	√	35619875
	4	系统性毒性	√	√	30717554
	5	心脏毒性	×	×	-
附子	1	心脏毒性	√	√	30261585
	2	肾毒性	×	×	-
	3	肝毒性	×	×	-
	4	系统性毒性	×	×	-
	5	其他毒性	×	×	-
鼻码子	1	肝毒性	√	√	12565756
	2	肾毒性	√	√	12565756
	3	其他毒性	√	√	16278363
	4	系统性毒性	√	√	16278363
	5	心脏毒性	√	√	16278363

4.3. HerbToxNet 能够发现中性属性草药中的毒性风险

为了检验传统假设，即“平性”（中性药性）草药本质上无毒，我们选择了白蔓子作为代表性平性草药，进行深入的个案分析。根据传统中医学理论，具有中性药性的草药被认为是平衡且安全的。如表S3所示，尽管在《本草纲目》等经典中医文献中被归类为中性草药，中性药通常被认为具有平衡特性且安全，毒性低

make accurate and reliable toxicity prediction of extremely “cold” or “hot” herbs.

Table S3: Results of toxicity prediction for *Chai Hu*, *He Shou Wu*, *Ba Dou*, *Fu Zi*, *Bi Ma Zi*

Herb Name	Rank	Toxicity	Prediction	Known	PubMed ID
Chai Hu	1	Hepatotoxicity	√	√	21749378
	2	Nephrotoxicity	√	√	34309413
	3	Systemic Toxicity	√	√	21749378
	4	Other Toxicity	×	×	-
	5	Cardiotoxicity	×	×	-
He Shou Wu	1	Hepatotoxicity	√	√	25648693
	2	**Systemic Toxicity**	√	×	**19873791**
	3	Nephrotoxicity	√	√	32426606
	4	Other Toxicity	×	×	-
	5	Cardiotoxicity	×	×	-
Ba Dou	1	Other Toxicity	√	√	35619875
	2	Nephrotoxicity	√	√	32943947
	3	Hepatotoxicity	√	√	35619875
	4	Systemic Toxicity	√	√	30717554
	5	Cardiotoxicity	×	×	-
Fu Zi	1	Cardiotoxicity	√	√	30261585
	2	Nephrotoxicity	×	×	-
	3	Hepatotoxicity	×	×	-
	4	Systemic Toxicity	×	×	-
	5	Other Toxicity	×	×	-
Bi Ma Zi	1	Hepatotoxicity	√	√	12565756
	2	Nephrotoxicity	√	√	12565756
	3	Other Toxicity	√	√	16278363
	4	Systemic Toxicity	√	√	16278363
	5	Cardiotoxicity	√	√	16278363

4.3. `HerbToxNet` *can uncover toxicity risks in neutral-property herbs*

To examine the traditional assumption that “ping” (neutral-property) herbs are inherently non-toxic, we selected *Bi Ma Zi*, a representative ping herb, for in-depth case analysis. According to classical TCM theory, herbs with a neutral property are considered balanced and safe. As shown in Table S3, despite being classified as a neutral herb in classical TCM texts such as the “Bencao Gangmu”, where neutral-property herbs are traditionally considered to possess balanced qualities and safe with low tox-

现代毒理学研究显示了对于白马仔不同的图景。这种草药含有高毒性成分，主要是蓖麻毒素和蓖麻碱，摄入后可能引起严重的全身毒性反应。具体而言，蓖麻毒素表现出明显的肝毒性，可导致肝细胞坏死及肝酶升高；同时具有肾毒性，表现为急性肾小管损伤和肾功能衰竭。此外，蓖麻毒素还可能引起心脏毒性，表现为低血压和心律失常；神经毒性，包括头痛和意识改变；胃肠道毒性，如严重呕吐、腹泻和出血；以及显著的免疫过敏反应。基于这些发现，选择白马仔作为一种代表性中性药材，批判性地评估传统认为中性药材天生无毒的假设。其目的是揭示潜在的毒性风险，并提供科学证据，从而完善和补充传统分类。

4.4. *HerbToxNet*可以揭示草药特性与毒性之间的关系

为了研究HerbToxNet预测与传统中医理论之间的相关性，我们还调查了中药属性与毒性的关系。多种中药的毒性，包括附子、何首乌、巴豆、弥麻子和柴胡，与其药理特性密切相关。附子具有热性、辛味，并归心、肾、脾经，以其生物碱毒性而引起心脏毒性和神经毒性而闻名。何首乌以苦味、归肝肾经为特征，经常与肝毒性相关。巴豆具有极烈的辛热特性，归胃大肠经，导致强烈的胃肠和全身毒性。虽然弥麻子传统上被认为性平、味甘，但其肝肾归经对应显著的肝毒性和肾毒性作用。柴胡为寒苦药，归肝经，也与肝毒性相关，考虑到...

icity, modern toxicological studies reveal a different picture for *Bi Ma Zi*. This herb contains highly toxic ingredients, primarily *ricin* and *ricinine*, which can induce severe systemic toxic effects upon ingestion. Specifically, *ricin* exhibits pronounced hepatotoxicity, leading to hepatocellular necrosis and elevated liver enzymes, and nephrotoxicity, characterized by acute tubular injury and renal failure [15, 16]. Additionally, *ricin* can cause cardiotoxicity, manifesting as hypotension and arrhythmias; neurotoxicity, including headaches and altered consciousness; gastrointestinal toxicity, such as severe vomiting, diarrhea, and hemorrhage; and significant immunoallergic reactions [16]. Based on these findings, *Bi Ma Zi* is selected as a representative neutral herb to critically assess the traditional assumption that neutral-property herbs are inherently non-toxic. The aim is to uncover potential toxic risks and to provide scientific evidences that may refine and complement the classical TCM classification system.

4.4. `HerbToxNet` *can uncover the relationship between herb properties and toxicity*

To examine the correlation between predictions of `HerbToxNet` and traditional TCM theory, we also investigate the relationship between TCM properties and toxicity. The toxicity of several herbs, including *Fu Zi*, *He Shou Wu*, *Ba Dou*, *Bi Ma Zi*, and *Chai Hu*, are closely associated with their pharmacological properties. *Fu Zi*, with its hot nature, acrid taste, and heart, kidney, and spleen tropism, is known for cardiotoxicity and neurotoxicity due to toxic alkaloids. *He Shou Wu*, characterized by bitter flavor and liver and kidney tropism, is frequently linked to hepatotoxicity. *Ba Dou*, bearing extreme acrid-hot properties and stomach and large intestine tropism, causes strong gastrointestinal and systemic toxicity. Although *Bi Ma Zi* is traditionally considered neutral and sweet, its liver and kidney meridian affinity corresponds with significant hepatotoxic and nephrotoxic effects. *Chai Hu*, a cold and bitter herb targeting the liver meridian, is also associated with liver toxicity, consistent with its traditional classification. These cases demonstrate that the core TCM properties of nature, flavor, and meridian tropism provide valuable indicators for assessing herb toxicity risk within the TCM paradigm. The toxicity predictions generated by `HerbToxNet` for these canonical herbs are consistent with the toxicity profiles. This suggests that `HerbToxNet` can effectively leverage the information encoded in traditional herb properties (nature,

（味、归经）用于预测草药毒性。

5. 生物可解读性分析

在预测草药毒性的任务中，草药成分与其作用靶点之间的相互作用在调节药理作用和毒性反应中起关键作用。单一草药通常含有多种生物活性成分，这些成分可以作用于不同的靶点，触发一系列信号事件，影响细胞功能和器官系统，最终表现为复杂的毒理表型。为了提升HerbToxNet的生物可解释性，我们构建了一个多层次解释框架，涵盖“草药–成分–靶点–生物通路–毒性”轴，并以柴胡作为代表案例进行生物学验证。

基于“模型可解释性分析”小节，我们首先通过HIH和HTH元路径识别了与柴胡相关的草药。随后，我们对这些草药进行了进一步分析。在成分方面，我们发现这些草药（由HIH元路径确认）共享某些类的次级代谢物，包括三萜皂苷、类黄酮、酚类化合物、挥发油和苦味生物碱。通过整合已知的毒性信息，我们最终确定了四种有毒成分，分别为柴胡皂苷A、柴胡皂苷D、黄芩素和丁香酚。我们从由HTH元路径确认的草药中提取了49个与肝毒性、肾毒性和全身毒性相关的独特靶点。这些靶点可能参与由已识别的有毒成分引发的毒性机制。随后，为了建立成分-靶点-毒性关联，我们进行了广泛的文献回顾和综述。

如表 S4 所示，从柴胡中鉴定出的有毒成分呈现出异质的毒理学特性，这反映在它们多样的靶点谱和与多种毒性类型的关联上。柴胡皂苷 A 和柴胡皂苷 D 是两种三萜皂苷，它们与多种凋亡和增殖相关的作用有关

flavor, meridian tropism) to predict herb toxicity.

5. Biointerpretability analysis

In the task of predicting herb toxicity, the interactions between herb ingredients and their targets play a pivotal role in mediating pharmacological effects and toxic responses. A single herb typically contains multiple bioactive ingredients that can act on diverse targets, triggering cascades of signaling events that influence cellular functions and organ systems, ultimately manifesting as complex toxicological phenotypes. To enhance the biointerpretability of `HerbToxNet`, we constructed a multi-level explanatory framework encompassing the “herb–ingredient–target–biological pathway–toxicity” axis, using *Chai Hu* as a representative case study for biological validation.

Based on the “Model interpretability analysis” subsection, we first identified the herbs associated with *Chai Hu* through the HIH and HTH meta-paths. We then conducted further analysis on these herbs. In terms of ingredients, we found that these herbs, as confirmed by HIH meta-path, share common classes of secondary metabolites, including triterpenoid saponins, flavonoids, phenolic compounds, volatile oils, and bitter alkaloids. By integrating known toxicity information, we ultimately identified four toxic ingredients, denoted as *Saikosaponin A*, *Saikosaponin D*, *Wogonin*, and *Eugenol*. We extracted 49 distinct targets that are implicated in hepatotoxicity, nephrotoxicity, and systemic toxicity from the herbs confirmed by HTH meta-path. These targets are potentially involved in toxicity mechanisms triggered by interactions with the identified toxic ingredients. Subsequently, to establish the ingredient–target–toxicity associations, we conducted an extensive literature review and cross-referenced the results with existing toxicity databases. As a result, we constructed a tripartite association table centered on the toxic ingredients, linking them to their corresponding targets and toxicity types.

As shown in Table S4, the toxic ingredients identified from *Chai Hu* display heterogeneous toxicological characteristics, as reflected by their diverse target profiles and associations with various toxicity types. *Saikosaponin A* and *Saikosaponin D*, two triterpenoid saponins, are associated with several apoptosis and proliferation related

表 S4：柴胡成分–靶点–毒性关系

成分类别	成分名称	靶标（基因符号）	毒性类型
三萜皂苷	柴胡皂苷A	CASP3、RB1、BCL2、BAX、MYC、CDKN2A	肝毒性，肾毒性
三萜皂苷	柴胡皂苷D	JUN, TGFB1, TNF, RELA, IL6, NR3C1, BCL2, NFKB2A, MYC, TP53, CDK4	肝毒性，肾毒性
类黄酮	黄酮醇	MAOA、MAOB、NOAS2、PTGS2、ABCC2、CYP1B1	全身毒性
酚类化合物	丁香酚	NFKB1、BAX、CASP3、TP53、PTGS2、BCL2、NOS2、CSK3B、VEGFA、CYP1A2、HIF1A、CTNNB1、CYP2C19、RELA、PRKCD、MMP1、MCL1、CCL2、CDK9、CCND1、BBC3、CKDN1A，IL6，CXCL8，KDR，AKT1，PTGS1，JUN，MMP9，CDKM1B，MAPK11，TNF，ERN1，EGFR，SPP1，CDK4，TRAF2，MAPK1	肝毒性

靶点如 CASP3、BCL2、MYC 和 TP53，提示其可能干扰肝脏和肾脏组织的细胞存活和稳态，从而导致肝毒性和肾毒性。黄酮类化合物黄酮醇可以作用于包括 MAOA、PTGS2 和 CYP1B1 在内的靶点，这些靶点已知参与炎症反应和代谢活化，表明其系统性毒性机制可能通过氧化应激和免疫失调介导。酚类化合物丁香酚显示出最广泛的靶点谱，共涉及 38 个不同靶点，如 IL6、TNF、NFKB1、EGFR 和 MAPK1，其中许多靶点在炎症信号传导、细胞应激反应和组织损伤中起核心作用。这种广泛的靶点参与强调了丁香酚与肝毒性的强烈关联，并凸显其作为柴胡中主要毒性成分的潜在作用。

为了进一步阐明生物学功能及相关的毒理学途径，我们对这些与毒性相关的靶点进行了KEGG通路富集分析和基因本体（GO）注释分析。如图S4所示，KEGG通路富集分析显示，显著富集的主要通路包括AGE-RAGE通路、巨细胞病毒感染、KSHV感染，以及一系列与癌症相关的通路，如癌症通路、胰腺癌和膀胱癌等。许多这些通路与慢性炎症、氧化应激以及已知的组织损伤和器官毒性的异常凋亡过程密切相关。例如，AGE-RAGE轴的激活已知可诱导活性氧(ROS)生成和NFKB激活，从而导致持续的炎症反应和纤维化重塑，这是其关键的病理特征。

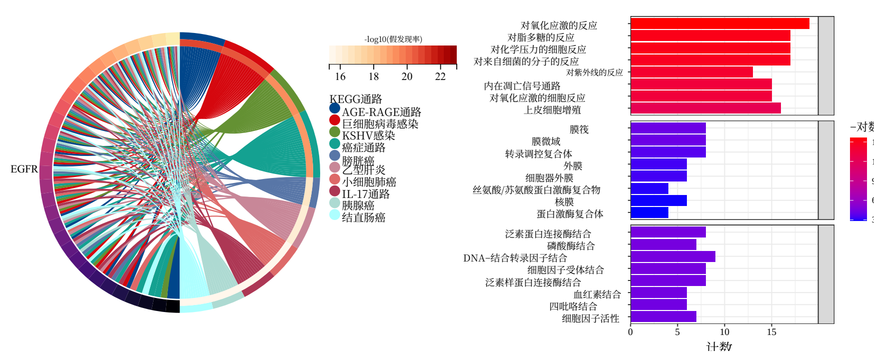
Table S4: *Chai Hu* ingredients–targets–toxicity relationship

Ingredient Class	Ingredient Name	Target (Gene Symbol)	Toxicity Type
Triterpenoid saponin	Saikosaponin A	CASP3, RB1, BCL2, BAX, MYC, CDKN2A	Hepatotoxicity, Nephrotoxicity
Triterpenoid saponin	Saikosaponin D	JUN, TGFB1, TNF, RELA, IL6, NR3C1, BCL2, NFKB2A, MYC, TP53, CDK4	Hepatotoxicity, Nephrotoxicity
Flavonoid	Wogonin	MAOA, MAOB, NOAS2, PTGS2, ABCC2, CYP1B1	Systemic toxicity
Phenolic compound	Eugenol	NFKB1, BAX, CASP3, TP53, PTGS2, BCL2, NOS2, CSK3B, VEGFA, CYP1A2, HIF1A, CTNNB1, CYP2C19, RELA, PRKCD, MMP1, MCL1, CCL2, CDK9, CCND1, BBC3, CKDN1A, IL6, CXCL8, KDR, AKT1, PTGS1, JUN, MMP9, CDKM1B, MAPK11, TNF, ERN1, EGFR, SPP1, CDK4, TRAF2, MAPK1	Hepatotoxicity

targets, such as *CASP3*, *BCL2*, *MYC*, and *TP53*, suggesting their potential to disrupt cell survival and homeostasis in hepatic and renal tissues, thereby contributing to both hepatotoxicity and nephrotoxicity. *Wogonin*, classified as a flavonoid, interacts with targets including *MAOA*, *PTGS2*, and *CYP1B1*, which are known to participate in inflammatory responses and metabolic activation, indicating a mechanism of systemic toxicity likely mediated by oxidative stress and immune dysregulation. *Eugenol*, a phenolic compound, exhibits the broadest target spectrum, with 38 distinct targets implicated, such as *IL6*, *TNF*, *NFKB1*, *EGFR*, and *MAPK1*, many of which are central to inflammatory signaling, cellular stress responses, and tissue injury. This extensive target involvement underscores *Eugenol*'s strong association with hepatotoxicity and highlights its potential role as a major toxic ingredient in *Chai Hu*.

To further elucidate the biological functions and involved toxicological pathways, we performed KEGG pathway enrichment and Gene Ontology (GO) annotation analyses on these toxicity-associated targets. As shown in Figure S4, KEGG pathway enrichment analysis revealed that the top significantly enriched pathways include the AGE-RAGE pathway, Cytomegalovirus infection, KSHV infection, and a range of cancer-associated pathways such as cancer pathways, pancreatic cancer, and bladder cancer. Many of these pathways are tightly linked to chronic inflammation, oxidative stress, and dysregulated apoptosis processes that are known mediators of tissue injury and organ toxicity. For example, the activation of AGE-RAGE axis is known to induce reactive oxygen species (ROS) production and NFKB activation, leading to sustained inflammatory responses and fibrotic remodeling, which are key pathological features

肝脏和肾脏损伤 [17]。同样，IL-17 信号通路和乙型肝炎途径的富集提示免疫驱动的细胞毒性和促炎性细胞因子释放，这两者都与肝脏炎症和肾脏炎症相关 [18]。

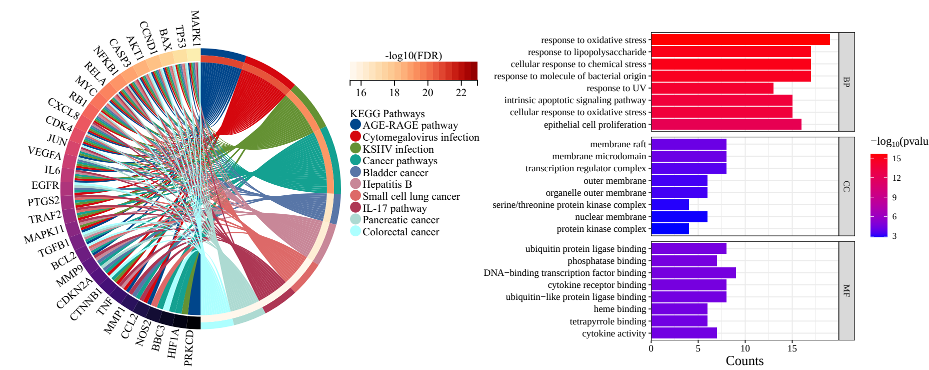


(a) KEGG 通路富集 (b) GO 注释

图 S4：毒性相关靶点的富集分析结果，（a）KEGG 通路富集，（b）基因本体注释。

GO 注释结果也印证了这些发现。在生物过程（BP）类别中，显著富集的术语如对氧化应激的反应、细胞对化学应激的反应、对脂多糖的反应以及内源性凋亡信号通路表明，所鉴定的靶点在细胞对有毒损伤的反应中起重要作用。这些生物过程代表了肝细胞和肾小管细胞损伤的核心机制，其中氧化应激和线粒体功能障碍常导致细胞凋亡或坏死。像上皮细胞增殖这样的富集术语可能反映了组织损伤后的代偿性再生。在细胞组分（CC）类别中，膜筏和细胞器外膜的富集提示有毒靶蛋白定位于参与信号转导和应激感知的关键亚细胞区室。富集的分子功能（MF）术语，如细胞因子受体结合，

of liver and kidney injury [17]. Similarly, IL-17 signaling and hepatitis B pathway enrichment suggest immune-driven cytotoxicity and pro-inflammatory cytokine release, both of which have been implicated in hepatic inflammation and nephroinflammation [18].



(a) KEGG pathway enrichment (b) GO annotation

Figure S4: Enrichment analysis results of toxicity-associated targets, (a) KEGG pathway enrichment, (b) Gene Ontology annotation.

GO annotation results also corroborate these findings. In the Biological Process (BP) category, significantly enriched terms such as response to oxidative stress, cellular response to chemical stress, response to lipopolysaccharide, and intrinsic apoptotic signaling pathway, indicate that the identified targets are heavily involved in the cellular response to toxic insults. These biological processes represent core mechanisms of hepatocyte and renal tubular cell injury, where oxidative stress and mitochondrial dysfunction often lead to apoptosis or necrosis. The enriched terms like epithelial cell proliferation may reflect compensatory regeneration following tissue injury. In the Cellular Component (CC) category, the enrichment of membrane raft and organelle outer membrane suggests the localization of toxic target proteins at key subcellular compartments involved in signal transduction and stress sensing. Enriched Molecular Function (MF) terms, such as cytokine receptor binding, ubiquitin protein ligase binding, and heme binding, are functionally related to inflammatory signaling, protein degradation, and metabolic enzyme activity, respectively. Notably, heme binding and tetrapyrrole binding are relevant to cytochrome P450 activity, implicating im-

异种生物代谢配对被认为是药物引起肝脏和肾脏损伤的一个因素。这些结果表明，柴胡的毒性作用靶点嵌入在一个复杂的信号网络中，该网络调控免疫激活、氧化应激反应、凋亡和代谢解毒。柴胡中的有毒成分可能干扰这些靶点，从而破坏关键的细胞过程，并最终导致肝毒性、肾毒性和系统性毒性。

总的来说，这项生物可解释性分析突显了 HerbToxNet 在揭示草药诱导毒性相关的机制性意义和生物学上合理的途径方面的能力。通过整合药理学和毒理学证据以及“草药–成分–靶点–毒性”轴，HerbToxNet 不仅增强了模型的可解释性，还为草药的风险评估提供了宝贵依据。柴胡的案例研究展示了如何系统地将多层次证据联系起来，以阐明潜在的毒理机制，从而支持更安全和更科学的草药使用。

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paired xenobiotic metabolism as a contributor to drug-induced liver and kidney damage. These results suggest that the toxic targets of *Chai Hu* are embedded in a complex signaling network that governs immune activation, oxidative stress responses, apoptosis, and metabolic detoxification. Toxic ingredients in *Chai Hu* may interfere with these targets, thereby disrupting critical cellular processes and ultimately contributing to hepatotoxicity, nephrotoxicity, and systemic toxicity.

In summary, this biointerpretability analysis highlights the capability of `HerbToxNet` to uncover mechanistically meaningful and biologically plausible pathways underlying herb-induced toxicity. By integrating pharmacological and toxicological evidence along with the “herb–ingredient–target–toxicity” axis, `HerbToxNet` not only enhances the model interpretability but also provides a valuable basis for risk assessment of herbs. This case study of *Chai Hu* exemplifies how multi-level evidences can be systematically linked to elucidate potential toxicological mechanisms, thereby supporting safer and more informed use of herbs.

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